MASTER PROJECT

Shock propagation through the lens of remittances

Who bears the cost? Evidence from US–Mexico migration corridors

Authors

Anqi Wang Liu, Francesco Zucca, Joan Costa Rodríguez,

Michael Fehl, Xènia Alemany Rovira

Barcelona School of Economics

Master's program in Economics

Academic year 2025–2026

Contents

Abstract

1. Introduction

In 2024, Mexico received more than 67 billion current US dollars in remittances, making it the second-largest recipient country in the world after India (World Bank, 2024). This figures represents approximately 3.7% of Mexico's GDP (reference?), highlighting the macroeconomic significance of these transfers. With roughly 96% of remittances to Mexico originating from the United States (reference?), and the average Mexican household in the US sending approximately US$12,800 per year (Isidro Luna & Martínez Hernández, 2026), a natural question arises: does an economic shock in the US propagate to Mexico through the remittance channel? Specifically, do Mexican migrants absorb the shock and maintain their transfer levels constant, or do they reduce remittance outflows in response to a decline in their current income or wealth? The answer has direct implications for understanding the vulnerability of Mexican households to external shocks and the extent to which the remittance channel acts as a shock transmission mechanism.

The closest paper to ours is Caballero et al. (2023), who study how the Great Recession affected remittance receipt among Mexican households. Using a shift-share research design that exploits variation in migration networks across Mexican municipalities, they find that households more connected to severely hit U.S. labor markets experiencing larger employment declines were significantly less likely to receive remittances. This confirms that the remittance channel acts as a transmission mechanism for US labor demand shocks. However, their analysis is limited to the extensive margin (i.e. whether the households receive remittances at all), while not providing any estimate for the intensive margin (i.e. how much remittance flows decline in response to the shock). Quantifying this magnitude is central to our paper, as it determines the actual income loss borne by Mexican households. Moreover, the Great Recession was a global shock of exceptional magnitude, which simultaneusly contracted both the U.S. and Mexican economies, with Mexico recording a 6.6 points decline in real GDP in 2009 (Villarreal, 2010). This makes it difficult to isolate the effect of the U.S. economic decline on Mexican remittance inflows. Furthermore, shocks of this magnitude are uncommon, which limits the external validity of the findings derived from them.

We address these challenges by exploiting the localized and more frequent nature of natural disasters. From 1980 to 2024, the U.S. has sustained 403 weather and climate disasters for which the individual damage costs reached or exceeded US$1 billion, with cumulative costs that exceeds US$2.915 trillion (NOAA National Centers for Environmental Information, 2025). These events generate localized reductions in firm productivity, leading to lower family income (Boustan et al., 2020) and therefore plausibly reducing the capacity of affected migrant households to remit.

The relationship between natural disasters and remittances has been studied before, but existing work focuses on the opposite side of the corridor, analyzing what happens to bilateral remittance flows when the *receiving* country is struck. Bettin et al. (2023) study the evolution of remittance flows from 103 Italian provinces to 79 developing countries and find that remittances increase by approximately 1.8% on impact following a disaster in the recipient country, rising to about 2.4% four months later and rapidly decreasing after one year. This provides a useful benchmark for the speed and persistence of remittance adjustment, although it operates through a different mechanism than ours, where the shock originates on the *sender* side.

We contribute to this literature by providing causal evidence on how a localized US shock propagates to Mexico through the remittance channel. To this end, we exploit Hurricane Ian (September 2022), the costliest hurricane in Florida's history, which caused US$ 109.5 billion in damage (Bucci et al., 2026), as a natural experiment. The storm struck the southwestern coast of Florida but did not affect Mexico, providing a clean one-sided shock to the sender side of the remittance corridor. We estimate bilateral remittance flows between each US state and each Mexican municipality by applying a migration weighting matrix to Banxico data on remittance inflows by U.S. State of origin (Banco de México, CE168, 2026).

Then, we implement a difference-in-differences design comparing Florida-linked municipalities against those connected to unaffected states. We find that ... Our findings suggest that ..., implying that ...

2. Theoretical framework and estimation technique

2.1. Remittance model

We introduce the remittance model.

2.2. Examining spillover effects

2.2.1. Geographical spillovers

2.2.2. Migration network spillovers

2.3. Estimation equation

Then, talk about the estimation technique: for sure DiD, but which type of DiD in particular? More than one type?

3. Data and bilateral remittance flow estimation

Bilateral remittance flow data between US states and Mexican municipalities is not publicly available. We estimate these flows by distributing state-level remittance outflows from the US to Mexican municipalities according to observed migration shares. The estimated inflows at the municipality level are then compared against official data to validate the accuracy of our procedure.

3.1. Data sources

The source of remittance data is Banco de México (Banxico), which provides two complementary datasets through its Sistema de Información Económica (SIE). The first, series CE168, records remittance inflows from each US state to Mexico in millions of current US dollars at quarterly frequency, covering the period from Q1 2013 to Q1 2026 (Banco de México, CE168, 2026). This series serves as the US state ourflows (r_s) for our bilateral flow estimation. The second, series CE166, records remittance inflows received by each Mexican municipality, also in millions of current US dollars at quarterly frequency and over the same sample period (Banco de México, CE166, 2026). This series acts as abenchmark to validate the accuracy of our estimates at an aggregate level.

To bridge these datasets, we utilize administrative records from the Matricula Consular de Alta Seguridad (MCAS) program, managed by the Instituto de los Mexicanos en el Exterior (IME) (Instituto de los Mexicanos en el Exterior, 2026). The publicly available version of this dataset provides a comprehensive registry of consular identification cards issued to Mexican nationals residing in the United States each year, from 2010 to 2024. While the MCAS does not constitute an exhaustive census of all migrants, it represents a robust proxy for the spatial distribution of Mexican migrants in the United States. In particular, it is highly representative of undocumented and recently arrived migrants, who face the strongest incentives to obtain consular identification (Caballero et al., 2018).

3.2. *Bilateral remittance estimation*

To estimate bilaterance remittance flows from U.S. states to Mexican municipalities, we construct a migration-based weighting matrix using annual MCAS from 2010 to 2024. Because these records capture new consular issuances rather than the complete stock of the remitting population, which also includes previously arrived migrants, we rely on the assumption of persistance of migration patterns over time. To mitigate the volatility of yearly migration fluctuation and construct a robust proxy for the long-term spatial distribution of Mexican migrants, we compute the average of these matrices across all years. We then apply this averaged weighting matrix to the state-level remittance outflows from Banxico's series CE168 to derive the U.S. to Mexican municipality bilateral flows.

Formally, we define $MCAS_{ms}$ as the average annual count of MCAS issuances to migrants from Mexican municipality m residing in U.S. state s. For each U.S. state, we calculate the average migration share, w_{ms}, which represents the proportion of that state s's Mexican population originating from the m municipality, as the ratio of $MCAS_{ms}$ to the total number of MCAS issuances from that state across all municipalities M:

$$w_{ms} = \frac{MCAS_{ms}}{\sum_{m=1}^{M} MCAS_{ms}} \tag{3.1}$$

By applying the weights computed from equation (3.1) to the official aggregate state-level outflows from Banxico (r_s), we derive the estimated bilateral remittance flow ($\hat{r}_{ms}$) from state s to municipality m:

$$\hat{r}_{ms} = r_s \cdot w_{ms} \tag{3.2}$$

This migration share estimation approach yields the final estimate of our outcome variable.

The validity of this estimation relies on two primary identification assumptions.

First, we assume the temporal persistence of migration origin-destination flows, implying that the distribution of migrants recorded in the 2010–2024 MCAS data is representative of the active remitting population. Second, we assume that, within a given U.S. state, the average amount remitted does not vary systematically by the migrant's municipality of origin. While this assumption ignores micro-level heterogeneity, it provides a baseline estimate of the bilateral remittance flows that can be validated against the aggregate data provided by Banxico' series CE166.

3.3. *Persistence assumption and estimate validation*

To ensure the robustness of the assumption discussed at the end of section 3.2, we conduct a validation exercise by aggregating our estimated municipal inflows and comparing them against the actual receipts recorded in Series CE166. Following the validation logic in Caballero et al. (2018), finding a high correlation between our estimates and the official municipal totals confirms that the weighting matrix accurately captures the underlying financial architecture of the US-Mexico remittance channel.

The estimated remittance inflow for each municipality is thus a weighted sum of the state-level outflows, where the weights are determined by the migration shares.

correlation of estimated remittances with real data on aggregate level

4. Shock selection and results

4.1. *Hurricane Ian, 2022*

The shock selected for this analysis is Hurricane Ian, which struck the southwestern coast of Florida on September 28, 2022, with peak sustained winds of 140 kt (around 260 kph) (Bucci et al., 2026). Ian caused an estimated US$ 112.9 billion in total damage in the United States, of which US$ 109.5 billion occurred in Florida alone, making it simultaneously the costliest hurricane in Florida's history and the third-costliest in US history. In Lee County alone, at least 52,514 structures were impacted, of which 5,369 were completely destroyed (Bucci et al., 2026).

Crucially for our identification strategy, Hurricane Ian did not affect Mexico. The storm originated in the Caribbean, made landfall in Cuba and Florida, and dissipated over the Atlantic after a final less impactful landfall in South Carolina (Bucci et al., 2026). Therefore, the shock is spatially clean. Indeed, remittance senders in Florida were directly exposed to a severe and sudden wealth shock (reference or data?), while remittance receivers in Mexican municipalities were entirely unaffected. This allows us to better isolate the transmission of the shock through the remittance channel.

4.2. *Preliminary statistics*

check parallel trends

statistics before (e.g. average over the past year), 1, 2, 3, and 4 quarters after the shock. I would do this with growth rates even if the estimation is in levels because quarterly growth rates are more interpretable and take into account the trend. Note that there is also seasonality to take into account.

4.3. DiD results

4.3.1. Benchmark model

4.3.2. Testing for spillover effects

5. Conclusion

Bibliography

Banco de México, CE166. (2026). *Ingresos por remesas, distribución por municipio (CE166)* [Sistema de Información Económica (SIE), Banco de México]. Retrieved May 7, 2026, from https://www.banxico.org.mx/SieInternet/consultarDirectorioInternetAction.do?accion=consultarCuadro&idCuadro=CE166&locale=es

Banco de México, CE168. (2026). *Estado de origen de los ingresos por remesas provenientes de Estados Unidos (CE168)* [Sistema de Información Económica (SIE), Banco de México. Trimestral, millones de dólares, flujos. Periodo: Ene-Mar 2013 – Jul-Sep 2024]. Retrieved May 7, 2026, from https://www.banxico.org.mx/SieInternet/consultarDirectorioInternetAction.do?accion=consultarCuadro&idCuadro=CE168&locale=es

Bettin, G., Jallow, A., & Zazzaro, A. (2023). *How do monthly remittances respond to natural disasters in migrants' home countries?* (Mo.Fi.R. Working Papers No. 179). Money, Finance Research group (Mo.Fi.R.) - Univ. Politecnica Marche - Dept. Economic, and Social Sciences. https://doi.org/None

Boustan, L. P., Kahn, M. E., Rhode, P. W., & Yanguas, M. L. (2020). The effect of natural disasters on economic activity in us counties: A century of data. *Journal of Urban Economics*, *118*, 103257. https://doi.org/https://doi.org/10.1016/j.jue.2020.103257

Bucci, L., Alaka, L., Hagen, A., Delgado, S., & Beven, J. (2026). *Tropical cyclone report: Hurricane Ian (AL092022), 23–30 September 2022* (tech. rep.). National Hurricane Center, National Oceanic and Atmospheric Administration (NOAA). Retrieved May 7, 2026, from https://www.nhc.noaa.gov/data/tcr/AL092022_Ian.pdf

Caballero, M. E., Cadena, B. C., & Kovak, B. K. (2018). Measuring geographic migration patterns using matrículas consulares. *Demography*, *55*(3), 1119–1145. https://doi.org/10.1007/s13524-018-0675-6

Caballero, M. E., Cadena, B. C., & Kovak, B. K. (2023). The international transmission of local economic shocks through migrant networks. *Journal of*

International Economics, *145*, 103832. https://doi.org/https://doi.org/10.1016/j.jinteco.2023.103832

Instituto de los Mexicanos en el Exterior. (2026). Estadísticas: Información estadística de la población mexicana en el exterior [Accessed: 2026-04-01]. *Secretaría de Relaciones Exteriores (SRE)*.

Isidro Luna, V. M., & Martínez Hernández, F. A. (2026). Mexican households in the united states: Average income and remittances sent to mexico. *Panorama Económico*, *21*(43), 31–50. https://doi.org/10.29201/peipn.v21i43.228

NOAA National Centers for Environmental Information. (2025). U.S. billion-dollar weather and climate disasters [Retired May 2025; data coverage through December 2024]. https://doi.org/10.25921/stkw-7w73

Villarreal, M. A. (2010). *The Mexican economy after the Global Financial Crisis* (CRS Report No. R41402). Congressional Research Service, Library of Congress. Retrieved May 8, 2026, from https://apps.dtic.mil/sti/tr/pdf/ADA529130.pdf

World Bank. (2024). Personal remittances, received (current US$) [World Bank Open Data. Accessed: 2026-05-07]. https://data.worldbank.org/indicator/BX.TRF.PWKR.CD.DT